



First and last name	Serial number	Group

Answers must be brief and justified.

Exercise (01): (09 points). Let $G = \mathbb{R} - \{-3\}$. We define on G an internal composition law denoted $*$ by:

$$\forall x, y \in G, \quad x * y = xy + 3(x + y) + 6.$$

1. Show that $(G, *)$ is an abelian group.

1. Associativity 1

$$\begin{aligned} \forall x, y, z \in G: (x * y) * z &= x * (y * z) \\ (x * y) * z &= (xy + 3(x + y) + 6) * z \\ &= (xy + 3(x + y) + 6)z + 3(xy + 3(x + y) + 6 + z) + 6 \\ &= xyz + 3xz + 3yz + 3xy + 9x + 9y + 9z + 24 \quad (1) \end{aligned}$$

$$\begin{aligned} x * (y * z) &= x * (yz + 3(y + z) + 6) \\ &= x(yz + 3(y + z) + 6) + 3(x + yz + 3(y + z) + 6) + 6 \\ &= xyz + 3xz + 3yz + 3xy + 9x + 9y + 9z + 24 \quad (2) \end{aligned}$$

(1) = (2) then $*$ is associative

2. Commutativity

$$\begin{aligned} \forall x, y \in G: x * y &= y * x \\ x * y &= xy + 3(x + y) + 6 \\ &= yx + 3(y + x) + 6 = y * x \end{aligned}$$

then $*$ is commutative.

3. The neutral element

$$\exists e \in G, \forall x \in G: x * e = e * x = x$$

As the law $*$ is commutative we take just one equation (1)

$$x * e = x \Rightarrow xe + 3(x + e) + 6 = x$$

$$\Rightarrow e(x + 3) = -2(x + 3)$$

$$\Rightarrow \boxed{e = -2} \text{ because } x \neq -3$$

4. The inverse element

$$\begin{aligned} \forall x \in G: \exists x^{-1} \in G: x * x^{-1} &= x^{-1} * x = e \\ x * x^{-1} = e &\Rightarrow x x^{-1} + 3(x + x^{-1}) + 6 = -2 \\ \Rightarrow x^{-1}(x + 3) &= -8 - 3x \quad (1) \\ \Rightarrow x^{-1} &= \frac{-8 - 3x}{x + 3} \quad (x \neq -3) \end{aligned}$$

We have $x^{-1} \in G$ because if we take

$$x^{-1} = -3 \Rightarrow \frac{-8 - 3x}{x + 3} = -3$$

$$\Rightarrow 1 = 0 \text{ (impossible)}$$

2. Let $f: (G; *) \rightarrow (\mathbb{R}^*; \times)$ such that

$$f(x) = x + 3.$$

Show that f is an isomorphism of the group and give the expression of f^{-1} .

$$\begin{aligned} \forall x, y \in G: f(x * y) &= f(x) \times f(y) \\ f(x * y) &= xy + 3x + 3y + 9 = x(y+3) + 3(y+3) = (x+3)(y+3) = f(x) \times f(y) \end{aligned}$$

(1)

Then f is a morphism

rest prove that f is bijective, In fact

$$\forall y \in \mathbb{R}^*, \exists! x \in G: y = f(x)$$

Let $y \in \mathbb{R}^*$ we have $y = x + 3 \Rightarrow x = y - 3 \in G$
because $y \neq 0$

then f is bijective

(1)

f morphism + f bijective then f is an isomorphism

$$f^{-1}: \mathbb{R}^* \rightarrow G: f^{-1}(x) = x - 3$$

(1)

3. Calculate for all $n \in \mathbb{N}^*$ the quantity

$$x^{(n)} = \underbrace{x * \dots * x}_{n \text{ times}}, \text{ with } x^{(0)} = -2.$$

we have

$$\begin{aligned} f(x^{(n)}) &= f(x * x * \dots * x) \\ (\text{As } f \text{ is morphism}) &= \underbrace{f(x) \times f(x) \times \dots \times f(x)}_{n \text{ times}} \\ &= (f(x))^n \end{aligned}$$

(1)

$$\begin{aligned} \text{Then } x^{(n)} &= f^{-1}(f(x^{(n)})) = f^{-1}((x+3)^n) \\ &= (x+3)^n - 3. \end{aligned}$$

4. Show that $H =]-3, +\infty[$ is a sub-group of $(G, *)$.

1. $e = -2 \in H$?

$-2 \in]-3, +\infty[$

Then, $e \in H$

$$x^{-1} = -3 + \frac{1}{x+3}$$

we have $x+3 > 0 \Rightarrow \frac{1}{x+3} > 0$

So $-3 + \frac{1}{x+3} > -3$

therefore $x^{-1} \in H$

then H is a sub-group.

2. $\forall x, y \in H; x * y \in H$

We notice that

$$x * y = xy + 3x + 3y + 6 = (x+3)(y+3) - 3$$

we have $x > -3$ $x \in H$

and $y > -3$ $y \in H$

then $x+3 > 0 \wedge y+3 > 0$

that means:

$$(x+3)(y+3) > 0$$

$$\Rightarrow (x+3)(y+3) - 3 > -3$$

$$\Rightarrow x * y > -3$$

$$\Rightarrow x * y \in H$$

3. $\forall x \in H, x^{-1} \in H$.

We have $x^{-1} = \frac{-3x-8}{x+3}$

$$= \frac{-3(x+3)+1}{x+3}$$

Exercise (02): (03 points). Let f and g be two applications defined from $\mathbb{R}$ to $\mathbb{R}$. Using the quantifiers and logical connectors, write the following proposition, and its negation:

P_1 : "The maps of the functions f and g intersect at a single point in the interval $[-\pi, \pi]$."

$P_1: \exists! x \in [-\pi, \pi], f(x) = g(x)$

$\neg P_1: (\forall x \in [-\pi, \pi], f(x) \neq g(x)) \vee (\exists x_1, x_2 \in [-\pi, \pi] (x_1 \neq x_2) : f(x_1) = g(x_1) \wedge f(x_2) = g(x_2))$

Is the proposition P_1 true for the applications $f(x) = \frac{1}{2} + \frac{1}{2} \cos^2(x)$ and $g(x) = -\cos(x)$?

$$f(x) = g(x) \Rightarrow \frac{1}{2} + \frac{1}{2} \cos^2 x = -\cos x$$

$$\textcircled{1} \Rightarrow \cos^2 x + 2\cos x + 1 = 0 \Rightarrow (\cos x + 1)^2 = 0$$

$$\Rightarrow \cos x = -1 \Rightarrow x = -\pi \vee x = \pi$$

Then the proposition p_1 is false for the applications f and g .

Exercise (03): (02points). Prove the following proposition:

$$\forall (a, b) \in \mathbb{R}^2 : a^2 + b^2 = 1 \Rightarrow |a + b| \leq \sqrt{2}.$$

We know that $(a - b)^2 \geq 0 \Rightarrow a^2 + b^2 - 2ab \geq 0$

$$a^2 + b^2 = 1 \Rightarrow 1 - 2ab \geq 0 \Rightarrow 2ab \leq 1$$

$$(a + b)^2 = \underbrace{a^2 + b^2}_1 + \underbrace{2ab}_{\leq 1} \leq 2 \Rightarrow |a + b| \leq \sqrt{2}$$

$\textcircled{2}$

Exercise (04): (06points). Let f be an application from $\mathbb{R}$ to $\mathbb{R}$ defined by:

$$f(x) = \ln\left(\frac{e^x}{1 + e^x}\right).$$

1. Determine the following sets:

$$f([- \ln(3); 0]), \quad f^{-1}(\{1; -2\}) \text{ and } f^{-1}([-1; 1]).$$

$$\textcircled{0.5} f([- \ln(3); 0]) = \left\{ f(-\ln(3)), f(0) \right\} \quad \left| \quad f^{-1}(\{1, -2\}) = \left\{ f^{-1}(1), f^{-1}(-2) \right\} \right.$$

$$\textcircled{0.5} \begin{aligned} f(-\ln(3)) &= f\left(\ln\left(\frac{1}{3}\right)\right) = \ln\left(\frac{1}{4}\right) \\ f(0) &= \ln\left(\frac{1}{2}\right) \\ f([- \ln(3); 0]) &= \left\{ \ln\left(\frac{1}{4}\right), \ln\left(\frac{1}{2}\right) \right\} \end{aligned} \quad \left| \quad \begin{aligned} f^{-1}(1) &= \left\{ x \in \mathbb{R} / f(x) = 1 \right\} \\ f(x) = 1 &\Rightarrow \ln\left(\frac{e^x}{1 + e^x}\right) = 1 = \ln e \\ &\Rightarrow \frac{e^x}{1 + e^x} = e \end{aligned} \right.$$

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$$e^x = e(1+e^x) \Rightarrow e^x(1-e) = e$$

$$(0, V) \Rightarrow e^x = \frac{e}{1-e} < 0$$

$$\text{Then } f^{-1}(1) = \emptyset$$

$$f^{-1}(-2) = \{x \in \mathbb{R} : f(x) = -2\}$$

$$f(x) = -2 \Rightarrow \ln\left(\frac{e^x}{1+e^x}\right) = \ln e^{-2}$$

$$\Rightarrow \frac{e^x}{1+e^x} = e^{-2} \Rightarrow e^x e^2 = 1+e^x$$

$$\Rightarrow e^x(e^2-1) = 1$$

$$\Rightarrow e^x = \frac{1}{e^2-1} \quad (0, V)$$

$$\Rightarrow x = \ln\left(\frac{1}{e^2-1}\right)$$

$$f^{-1}(\{1, -2\}) = \left\{ \ln\left(\frac{1}{e^2-1}\right) \right\}$$

$$f^{-1}([-1, 1[) = \{x \in \mathbb{R} : -1 \leq f(x) < 1\}$$

$$-1 \leq \ln\left(\frac{e^x}{1+e^x}\right) < 1$$

$$\ln\left(\frac{e^x}{1+e^x}\right) < 1 \Rightarrow \frac{e^x}{1+e^x} < e$$

$$\Rightarrow e^x > \frac{e}{\frac{1-e}{<0}} \Rightarrow x \in \mathbb{R}$$

$$\ln\left(\frac{e^x}{1+e^x}\right) \geq -1 \Rightarrow \frac{e^x}{1+e^x} \geq \frac{e^{-1}}{1+e^{-1}}$$

$$\Rightarrow e^x(e-1) \geq 1 \Rightarrow e^x \geq \frac{1}{e-1}$$

$$\Rightarrow x \geq \ln\left(\frac{1}{e-1}\right)$$

$$\Rightarrow x \in \left[\ln\left(\frac{1}{e-1}\right), +\infty \right[$$

$$f^{-1}([-1, 1[) = \mathbb{R} \cap \left[\ln\left(\frac{1}{e-1}\right), +\infty \right[$$

$$(0, 78) = \left[\ln\left(\frac{1}{e-1}\right), +\infty \right[$$

2. Is the application f surjective on $\mathbb{R}$? Justify.

f is not surjective because $f^{-1}(1) = \emptyset$

that means for $y = 1$ the equation $f(x) = 1$

has not a solution. $(0, V)$

3. Show that f is injective on $\mathbb{R}$.

$$\forall x_1, x_2 \in \mathbb{R} : f(x_1) = f(x_2) \Rightarrow \ln\left(\frac{e^{x_1}}{e^{x_1}+1}\right) = \ln\left(\frac{e^{x_2}}{e^{x_2}+1}\right)$$

$$\Rightarrow \frac{e^{x_1}}{e^{x_1} + 1} = \frac{e^{x_2}}{e^{x_2} + 1} \Rightarrow e^{x_1} e^{x_2} + e^{x_1} = e^{x_1} e^{x_2} + e^{x_2}$$

$$\Rightarrow e^{x_1} = e^{x_2} \Rightarrow x_1 = x_2$$

Then f is injective.

(1)

4. Show that f is bijective from $\mathbb{R}$ to $] -\infty, 0[$ and give the inverse application f^{-1} .

f bijective $\Leftrightarrow \forall y \in] -\infty, 0[, \exists ! x \in \mathbb{R} : y = f(x)$

$$y = f(x) \Rightarrow y = \ln \left(\frac{e^x}{1 + e^x} \right) \Rightarrow e^y = \frac{e^x}{1 + e^x}$$

$$\Rightarrow e^y + e^x e^y = e^x \Rightarrow e^x (1 - e^y) = e^y$$

(1)

$$\Rightarrow e^x = \frac{e^y}{1 - e^y} \quad 1 - e^y > 0$$

$$\Rightarrow \ln \left(\frac{e^y}{1 - e^y} \right) = x$$

Then f is bijective.

$$f^{-1} :] -\infty, 0[\longrightarrow \mathbb{R}$$

$$x \longrightarrow \ln \left(\frac{e^x}{1 - e^x} \right)$$

$$f^{-1}(x) = \ln \left(\frac{e^x}{1 - e^x} \right)$$

0,78